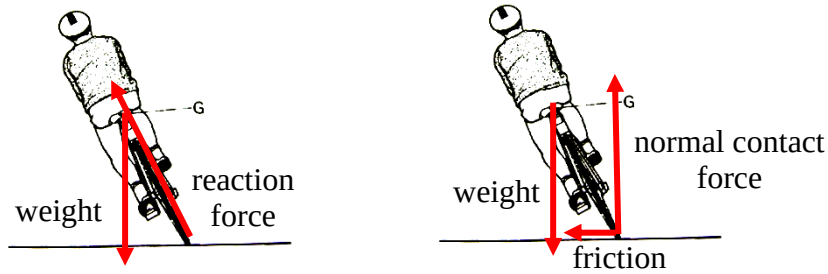


4 (a)



1 mark for correct direction of arrows

B1

1 mark for similar length of arrows in vertical direction

B1

(b) Since frictional force provides for the centripetal force,

$$f = \frac{mv^2}{R}$$

$$70 = \frac{(80)v^2}{55}$$

$$v = 6.9 \text{ m s}^{-1}$$

A1

(c) (i) For a surface which is banked, the horizontal component of the normal reaction is an additional source of the centripetal force.

B1

Since Centripetal Force = $\frac{mv^2}{r}$, an increase in the centripetal force

B1

would allow the rider to turn the corner at a higher speed without slipping, provide the mass of cyclist and bicycle and radius of turn remain constant.

(ii) Considering forces acting on the rider/bicycle in the horizontal direction:

$$N \sin 20^\circ + f \cos 20^\circ = \frac{mv^2}{r} \text{ ----- (1)}$$

C1

For the equilibrium in the vertical direction:

$$N \cos 20^\circ = mg + f \sin 20^\circ \text{ ----- (2)}$$

C1

Solving (1) & (2),

$$\tan 20^\circ (mg + f \sin 20^\circ) = \frac{mv^2}{r} - f \cos 20^\circ$$

A1

→ max velocity during turning $v = 15.7 \text{ m s}^{-1}$

[Turn over]

5	2	The rocket experiences equal and opposite gravitational forces exerted	P1
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